A log-concavity criterion for finite nonnegative linearization

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Abstract. In a 2005 survey, R. Szwarc asked for criteria, stated in the recurrence coefficients, for nonnegative linearization (property (P)) of finite orthogonal polynomial systems such as the Krawtchouk polynomials. For a finite three-term recurrence normalized to have row sums one, whose diagonal and downward coefficients have a constant nonnegative ratio, we prove that property (P) holds for every product, including products beyond the terminal degree, whenever the Newton coefficients of the profile $d_i = b_i + c_i$ are log-concave without internal zeros. The criterion contains the full positive Krawtchouk range and the dual q-Krawtchouk arrays, and through a quadratic decomposition of symmetric finite systems it covers Szwarc's quadratic system Q_n in its sharp range $0 < p_0 \leq 1/2$. Coefficient balls of a fixed radius around the Krawtchouk arrays cannot serve all sizes: for a fixed ratio $c_i/d_i = r < 1$, property (P) fails at coefficient distance $O((1-r^2)^N)$, an upper bound on the stability radius. All theorems, corollaries and propositions are formalized in Lean 4.

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1. Introduction

Nonnegative linearization (property (P)) asks whether the product of two orthogonal polynomials is a nonnegative linear combination of polynomials of the same family. For a finite system $R_0, \ldots, R_N$ the identity is understood on the orthogonality support, so it includes products whose degree sum exceeds N. In the normalization $R_i(1) = 1$ the coefficients then sum to one and define a finite polynomial hypergroup.

In his survey, Szwarc [20, §5, Problem 2] asks: "Find any criteria for the property (P) for finite systems of orthogonal polynomials, like the Krawtchouk polynomials."[1] He then

[1] In the author version, §5, Problem 2, printed p. 125, the displayed weights $\binom{N}{n}p^{N-n}(1-p)^n$ have mean $N(1-p)$, whereas the displayed recurrence gives $K_1(x) = (x-pN)/((1-p)N)$, so that $\int K_1\, d\mu = (1-2p)/(1-p)$. Corollary 3.6 fixes the convention directly: it uses the displayed weights and the normalization at the point carrying the weight p^N.

introduces a symmetric system Q_n built from the Krawtchouk recurrence, notes that it has property (P) for $p_0 \le 1/2$, and asks for a general criterion that would handle this case.

We work with a normalized three-term recurrence (2.2), whose coefficients have row sums one, with diagonal coefficients b_i, downward coefficients c_i and profile $d_i = b_i + c_i$. The main result, Theorem 3.1, concerns arrays with a constant ratio b_i/c_i: property (P) holds for every product whenever the Newton coefficients $\nu_k = \Delta^k d(0)$ of the profile are nonnegative and log-concave without internal zeros. This is a set of inequalities on the coefficients; checking it does not involve any product coefficient. It contains the full positive Krawtchouk range (Corollary 3.6; property (P) is classical there, see Eagleson [6], Dunkl and Ramirez [5] and [20, §5]) and the geometric arrays of Corollary 3.7, the dual q-Krawtchouk systems of Koelink's construction on the Hecke algebra of type B [8]. Section 4 answers the second part of the problem: a quadratic decomposition reduces property (P) of a symmetric finite system to two nonsymmetric ones (Theorem 4.1), and Theorem 3.1 then gives a parity-split criterion (Corollary 4.2) that covers Q_n for every N and $0 < p_0 \le 1/2$; this range is sharp (Theorem 4.3).

Section 5 discusses how far criteria of this kind can go. For every fixed $r < 1$, the radius $\rho_N(r)$ of the largest coefficient ball around $K_{N,r}$ in which every admissible array with $b_i \ge 0$ has property (P) is at most $3(1-r^2)^N$ for large N (Corollary 5.3). This is an upper bound only: the best lower bound we know, $\frac{1}{2}(\min(r, 1-r)/(32N))^N$ from [13], decays faster than every exponential, and the order of $\rho_N(r)$ is open. These statements concern fixed $r < 1$ and $N \to \infty$. In the other regime, fixed $N \ge 2$ and $r \to 1$, the radius $\rho_N(r)$ is of exact order $(1-r)^N$ [13, Corollary 9.11]; the two results are compatible. Still, no set of arrays with property (P) contains coefficient balls of a radius independent of N around the Krawtchouk arrays, so a criterion valid for all N must place them near its boundary; the log-concavity condition places them on its edge. Positivity of the Newton coefficients does not suffice, but log-concavity is not necessary; we record exact examples and pose the exact boundary as a problem.

Related work. Szwarc's criteria for infinite systems [20, Theorems 3.2 and 3.3] are monotonicity conditions on the recurrence coefficients; for a finite system some of them fail near the terminal index, whose upward coefficient vanishes (Section 4). Młotkowski and Szwarc [11] gave chain-sequence conditions for infinite systems whose orthogonality measure is supported on a convergent sequence of points, a case that the monotonicity criteria cannot reach. Askey [1, Part II, Theorem 1 and Corollary 1] gave finite recurrence conditions for products of degree sum at most N, through a maximum principle for a hyperbolic difference equation, and Szwarc [19, Theorem 1] characterized property (P) for finite and infinite systems through an auxiliary discrete hyperbolic problem. Miclo [9, Proposition 3.3] characterizes the hypergroup property of a finite birth–death chain at a boundary point spectrally. Having found numerically that two natural discrete versions of the log-concavity theorem of Achour and Trimèche fail, he concluded that "the right notion of log-concavity of a measure (or rather of a Markov kernel) has yet to be found in the discrete setting" [9, p. 433]. His partial discrete analogue [10, Theorem 31] gives inequality conditions for chains that are symmetric under $i \mapsto N - i$; the Krawtchouk systems with $p_0 \ne 1/2$ are not symmetric, and the symmetric one violates his condition that $2^i a_i$ be nonincreasing on $[0, N/2]$

once $N > 2$. None of these gives a coefficient criterion that covers all products of the finite Krawtchouk systems. For the finite systems of the Askey scheme a related question is open: Terwilliger [21, Problem 36.4] asks which real Leonard systems have only nonnegative intersection numbers. Blau and Hein [2, Theorem 1.1] determine the recurrences of the homogeneous monotonic P-polynomial table algebras, which are finite systems with nonnegative linearization in which all nontrivial elements have the same Haar weight and the recurrence coefficients are monotone.

What is new, imported and known. New here are Theorem 3.1, the quadratic decomposition and the parity-split criterion (Theorem 4.1 and Corollary 4.2), the coverage of Q_n by a coefficient criterion with the sharp range (Theorem 4.3), the upper bound on the stability radius (Corollary 5.3) and the examples on the sharpness of log-concavity. Nothing is imported: the written proofs do not use the companion papers, which are cited for comparison only, for instance [13] for a lower bound on the radius and for second proofs. Previously known are property (P) of the Krawtchouk arrays [6, 5, 20] and of the arrays $G_{N,q,r}$, which come from Koelink's construction [8] and, for some parameters, from distance-regular graphs; Szwarc states property (P) of Q_n for $p_0 \le 1/2$ [20, §5].

Formalization. All theorems, corollaries and propositions, as well as Lemmas 2.1 and 5.1, are formalized in Lean 4 with Mathlib; the lemmas of Section 3 are formalized within the proof of Theorem 3.1, whose Lean proof follows the written one. Some Lean proofs take another route, through the Lean development of [13]. Lean derives the positivity statements of Corollaries 3.6, 3.7 and 4.2, and through Corollary 4.2 that of Theorem 4.3, from the geometric-increment criterion of [13] rather than from Theorem 3.1. It proves Lemma 5.1, on which Proposition 5.2 and Corollary 5.3 rest, in the algebra of [13]. These Lean proofs verify the statements given here by routes different from the written proofs; details are in the supplement [16].

Two companion papers depend on this one: [12] treats ratios c_i/d_i that increase with the index, and [13] gives explicit perturbation neighborhoods, inequality criteria for finite systems away from the deepest products, a reversal method for palindromic systems and constraints at the symmetric Krawtchouk endpoint, with a finite-algebra proof for geometric increments as their reference positivity.

2. Setting

For integers $j \ge 0$ put $[j]_q = 1 + q + \cdots + q^{j-1}$, with $[0]_q = 0$; in particular $[j]_1 = j$. Consider real coefficients indexed by $0 \le i \le N$, where $N \ge 1$, satisfying

$$\begin{gathered} a_0 = 1, \quad b_0 = c_0 = 0, \quad a_N = 0, \quad a_i + b_i + c_i = 1, \\ a_i > 0 \ (i < N), \qquad c_i > 0 \ (i > 0). \end{gathered} \tag{2.1}$$

Set $R_{-1} = 0$, $R_0 = 1$, and determine $R_1, \dots, R_N$ by

$$xR_i = a_i R_{i+1} + b_i R_i + c_i R_{i-1} \qquad (0 \le i < N). \tag{2.2}$$

At $i = 0$, (2.2) gives $R_1 = x$; induction at $x = 1$, using the row sums and $a_i > 0$ for $i < N$, gives $R_i(1) = 1$ for $0 \le i \le N$. The terminal relation is $(x - b_N)R_N - c_N R_{N-1} = 0$ on

the orthogonality support. Linearization means equality of functions on this support, rather than an unrestricted polynomial identity when the degree sum exceeds N.

Write the tridiagonal recurrence matrix of such an array with the row convention

$$J_{i,i+1} = a_i, \qquad J_{i,i} = b_i, \qquad J_{i,i-1} = c_i,$$

with all other entries zero and indices outside $0, \dots, N$ omitted. We call an array satisfying (2.1) *normalized*. Its matrix J has row sums one, but (2.1) allows $b_i < 0$; we call the array *stochastic* only when all $b_i \geq 0$, so that J is a stochastic matrix. The arrays of Theorem 3.1 below are stochastic.

Lemma 2.1 *(Finite recurrence realization and triangle support).* *Every recurrence satisfying (2.1) has a positive orthogonality measure on exactly $N+1$ real points. If J is its tridiagonal recurrence matrix, then its finite product coefficients satisfy*

$$g(m,n;k) = [R_m(J)]_{n,k}. \tag{2.3}$$

They vanish unless $|m-n| \leq k \leq \min(m+n, N)$, regardless of their signs inside this interval.

Proof. Choose positive π_i with $\pi_0 = 1$ and $\pi_{i+1} = \pi_i a_i / c_{i+1}$. For $D = \operatorname{diag}(\sqrt{\pi_i})$, $H = DJD^{-1}$ is real symmetric. Induction in (2.2) gives the row-vector identity $e_0^{\mathsf{T}} R_i(J) = e_i^{\mathsf{T}}$, hence $e_0^{\mathsf{T}} R_i(H) = \pi_i^{-1/2} e_i^{\mathsf{T}}$. In particular e_0 is cyclic for H. In an orthogonal eigenspace decomposition of H, the cyclic span of e_0 has dimension at most the number of distinct eigenvalues; here it is all of $\mathbb{R}^{N+1}$, so H has $N+1$ distinct eigenvalues and the projection of e_0 on each eigenspace is nonzero. The spectral measure μ of H at e_0 therefore has positive mass at each of $N+1$ real points, and

$$\int R_i R_j \, d\mu = e_0^{\mathsf{T}} R_i(H) R_j(H) e_0 = \mathbf{1}_{i=j} \pi_i^{-1}.$$

The terminal relation holds on this support: its polynomial $Q(x) = (x - b_N) R_N - c_N R_{N-1}$ satisfies $e_0^{\mathsf{T}} Q(J) = 0$. Since e_0 is cyclic and $Q(J)$ commutes with J, $Q(J) = 0$. Multiplying a basis polynomial by x acts by the corresponding row of J, including the terminal row. Induction on powers of x therefore gives (2.3). Orthogonality also gives

$$g(m,n;k) = \frac{\int R_m R_n R_k \, d\mu}{\int R_k^2 \, d\mu}.$$

If one of m, n, k exceeds the sum of the other two, its polynomial is orthogonal to their product, whose degree is smaller. This proves the asserted support. ∎

We say that the array has *nonnegative linearization*, or *property (P)*, if $g(m,n;k) \geq 0$ for all $0 \leq m, n, k \leq N$. The coefficients $g(m,n;\cdot)$ always sum to one, since $R_i(1) = 1$ and 1 lies in the support, because $J\mathbf{1} = \mathbf{1}$.

For $q \geq 1$ and $0 < r \leq 1$ let $G_{N,q,r}$ be the array

$$a_i = \frac{q^i [N-i]_q}{[N]_q}, \qquad b_i = (1-r)\frac{[i]_q}{[N]_q}, \qquad c_i = r\frac{[i]_q}{[N]_q}, \tag{2.4}$$

with $[j]_q$ as above; it satisfies (2.1), since $[N]_q - [i]_q = q^i[N-i]_q$. Its profile $d_i = [i]_q/[N]_q$ has geometric increments with ratio q, and $b_i/c_i = (1-r)/r$ is constant. The Krawtchouk array is $K_{N,r} = G_{N,1,r}$, with $a_i = (N-i)/N$, $b_i = (1-r)i/N$ and $c_i = ri/N$.

3. Log-concave profiles

For an array satisfying (2.1) with profile $d_i = b_i + c_i$ write

$$\nu_k = \sum_{i=0}^{k} (-1)^{k-i} \binom{k}{i} d_i \quad (1 \le k \le N), \qquad d_n = \sum_{k=1}^{n} \binom{n}{k} \nu_k \quad (0 \le n \le N);$$

below we put $\nu_0 = 0$ and $\nu_k = 0$ for $k > N$. We say that $(\nu_k)_{1 \le k \le N}$ is *log-concave without internal zeros* if all $\nu_k \ge 0$, the indices with $\nu_k > 0$ form an interval $\{1, \dots, K\}$, and $\nu_k^2 \ge \nu_{k-1}\nu_{k+1}$ for $2 \le k \le N-1$. Note $\nu_1 = d_1 > 0$.

Theorem 3.1 *(Log-concave Newton criterion).* *Let $0 = d_0 < d_1 < \cdots < d_N = 1$ have Newton coefficients that are log-concave without internal zeros, let $0 < r \le 1$, and put $a_i = 1 - d_i$, $b_i = (1-r)d_i$, $c_i = rd_i$. Then the system has nonnegative linearization.*

The hypotheses are inequalities in the profile. They hold for the Krawtchouk arrays ($\nu = (1/N, 0, \dots, 0)$) and for all arrays $G_{N,q,r}$ (ν_k proportional to $(q-1)^{k-1}$, log-concave with equality). The hypotheses also hold for every admissible convex quadratic profile: $\nu_1 > 0$, $\nu_2 \ge 0$ when $N \ge 2$, and $\nu_k = 0$ for $3 \le k \le N$. The hypotheses exclude the known counterexamples with constant ratio: the convex counterexamples recorded in [13] have Newton coefficients of both signs, and for the combinations $(1-\lambda)K_{N,r} + \lambda G_{N,q,r}$ of Section 5, with $q > 1$, $N \ge 3$ and $0 < \lambda < 1$, one has $\nu_1 = (1-\lambda)/N + \lambda/[N]_q$ and $\nu_k = \lambda(q-1)^{k-1}/[N]_q$ for $2 \le k \le N$, so $\nu_2^2 - \nu_1\nu_3 = -\lambda(1-\lambda)(q-1)^2/(N[N]_q) < 0$.

Outline of the proof. The proof has four steps. (1) A rescaling removes r from the downward coefficients, and in the basis $E_i = e_i p_i$ the product coefficients become positive multiples of numbers L^k_{mn}. For fixed k these form a symmetric generating function $K_k(z, w)$, which an explicit difference operator annihilates; the operator includes the terminal row, so the argument also covers products of degree sum greater than N. Expanding $K_k = \sum \omega_{\ell,a}(z+w)^a(zw)^\ell$, it suffices to show $\omega_{\ell,a} \ge 0$. (2) Lemma 3.2 turns the annihilation into a recurrence that expresses ω at level $\ell + 1$ through lower levels, by *moves* with explicit weights $a_{\ell+a}$, tA_j, B_j and C_j; the moves of odd index j carry a minus sign. The recurrence holds for every profile, and its weights are nonnegative when all $\nu_k \ge 0$. (3) Log-concavity enters only through Lemmas 3.3 and 3.4: each move of odd index j is dominated by a quarter of the two-step paths through moves of the even indices 0 and $j-1$. (4) In operator form, $\omega = \beta + (\mathcal{U} + \mathcal{P} - \mathcal{Q})\omega$ with $\mathcal{P}^2 \ge 4\mathcal{Q}$, and induction on ℓ gives $\omega \ge \frac{1}{2}\mathcal{P}\omega \ge 0$.

A rescaled system and a kernel. Fix r and put $t = (1-r)/\sqrt{r} \ge 0$. If p_i is defined by $p_0 = 1$ and $yp_i = a_i p_{i+1} + t d_i p_i + d_i p_{i-1}$ $(0 \le i < N)$, then $R_i(x) = r^{i/2} p_i(x/\sqrt{r})$,

so the coefficients H^k_{mn} in $p_m p_n = \sum_k H^k_{mn} p_k$ on the support satisfy $g(m,n;k) = r^{(m+n-k)/2} H^k_{mn}$. It suffices to show $H^k_{mn} \geq 0$. Put

$$e_i \;=\; \prod_{j=0}^{i-1} \frac{a_j}{d_{j+1}}, \qquad E_i \;=\; e_i p_i.$$

Then $yE_i = d_{i+1}E_{i+1} + t d_i E_i + a_{i-1}E_{i-1}$ on the support for $0 \leq i \leq N$, with the terms $d_{N+1}E_{N+1}$ and $a_{-1}E_{-1}$ omitted; at $i = N$ this is the terminal relation. Write $E_m E_n = \sum_k L^k_{mn} E_k$, so that $H^k_{mn} = L^k_{mn} e_k/(e_m e_n)$, and

$$K_k(z,w) \;=\; \sum_{m,n=0}^{N} L^k_{mn} z^m w^n.$$

Let $\theta_z = z\,\partial_z$ and let $d(\theta) = \sum_k \nu_k \binom{\theta}{k}$ act on polynomials by $d(\theta_z) z^i = d_i z^i$. For $F(z) = \sum_{i=0}^N E_i z^i$ the recurrence reads $yF(z) = \mathcal{L}_z F(z)$ with

$$\mathcal{L}_z \;=\; (z^{-1} + t - z)\, d(\theta_z) + z,$$

because $d_0 = 0$ removes the term $z^{-1}E_0$ and $1 - d_N = 0$ removes the term $z^{N+1}E_N$. Applying this to $F(z)F(w) = \sum_k K_k(z,w) E_k$ in the two variables separately and comparing coefficients of the basis E_k gives $(\mathcal{L}_z - \mathcal{L}_w) K_k = 0$. Since K_k is symmetric, $K_k = \sum_{\ell,a\geq 0} \omega_{\ell,a} S^a Z^\ell$ with $S = z + w$, $Z = zw$, and $K_k(z,0) = z^k$ gives $\omega_{0,a} = \mathbf{1}_{a=k}$. As K_k has degree at most N in z, comparing the coefficients of $z^{\ell+a} w^\ell$ by induction on ℓ gives $\omega_{\ell,a} = 0$ whenever $\ell + a > N$.

The kernel recurrence. Put $f_n(\ell) = \sum_{h\geq 0} \binom{\ell}{h} \nu_{n+h}$ with $\nu_0 = 0$ and $\nu_k = 0$ for $k > N$, so $f_0(\ell) = d_\ell$, and

$$W_{p,q}(\ell, a) \;=\; \sum_{n \geq p} \binom{n-p+q}{q} \binom{a}{n} f_n(\ell) \;\geq\; 0,$$

and $A_j = W_{2j+1,j}$, $B_j = W_{2j+2,j}$, $C_j = W_{2j+2,j+1}$.

Lemma 3.2. *For all $\ell, a \geq 0$,*

$$\begin{aligned} d_{\ell+1}\omega_{\ell+1,a} \;=\; a_{\ell+a}\,\omega_{\ell,a} + \sum_{j\geq 0} (-1)^j \Big(& t\, A_j(\ell - j, a+2j+1)\,\omega_{\ell-j,a+2j+1} \\ & + B_j(\ell - j, a+2j+2)\,\omega_{\ell-j,a+2j+2} \\ & + C_j(\ell - j - 1, a+2j+2)\,\omega_{\ell-j-1,a+2j+2} \Big), \end{aligned} \tag{3.1}$$

where $a_i = 1 - d_i$ is extended by the polynomial d_i for $i > N$ and terms with a negative index are omitted.

Proof. Since $S^a Z^\ell = \sum_i \binom{a}{i} z^{\ell+i} w^{\ell+a-i}$, Vandermonde's identity $\binom{\ell+i}{k} = \sum_{p+q=k} \binom{\ell}{q}\binom{i}{p}$ and $\binom{a}{i}\binom{i}{p} = \binom{a}{p}\binom{a-p}{i-p}$ give

$$d(\theta_z)\, S^a Z^\ell \;=\; \sum_{p,q\geq 0} \nu_{p+q} \binom{\ell}{q} \binom{a}{p} z^p S^{a-p} Z^\ell.$$

Hence $(\mathcal{L}_z - \mathcal{L}_w)S^aZ^\ell$ equals $(z-w)S^aZ^\ell$ plus $\sum_{p,q}\nu_{p+q}\binom{\ell}{q}\binom{a}{p}S^{a-p}Z^\ell$ times $(z^{p-1} - w^{p-1}) + t(z^p - w^p) - (z^{p+1} - w^{p+1})$. Divide by $z-w$, using $\chi_n = (z^{n+1} - w^{n+1})/(z-w) = \sum_{0\le j\le\lfloor n/2\rfloor}(-1)^j\binom{n-j}{j}S^{n-2j}Z^j$ for $n \ge 0$, $\chi_{-1} = 0$ and $(z^{-1} - w^{-1})/(z-w) = -Z^{-1}$. Since $\sum_q \binom{\ell}{q}\nu_{p+q} = f_p(\ell)$, the quotient is

$$
\begin{aligned}
&- d_\ell S^aZ^{\ell-1} + a_{\ell+a}S^aZ^\ell \\
&+ \sum_{j\ge0}(-1)^j\Big(tA_j(\ell,a)S^{a-2j-1}Z^{\ell+j} + B_j(\ell,a)S^{a-2j-2}Z^{\ell+j} + C_j(\ell,a)S^{a-2j-2}Z^{\ell+j+1}\Big).
\end{aligned}
$$

Here $-d_\ell S^aZ^{\ell-1}$ comes from $p=0$; the term $a_{\ell+a}S^aZ^\ell$ combines S^aZ^ℓ with the parts $j=0$ of $-\chi_p$, which sum to $-d_{\ell+a}S^aZ^\ell$ by Vandermonde; and the three families from χ_{p-2}, $t\chi_{p-1}$ and $-(\chi_p - S^p)$, whose coefficients $\binom{p-2-j}{j}$, $\binom{p-1-j}{j}$ and $\binom{p-1-j}{j+1}$ are those of B_j, A_j and C_j. Summing over ℓ, a with weights $\omega_{\ell,a}$ and taking the coefficient of S^aZ^ℓ gives (3.1), because the monomials S^aZ^ℓ are linearly independent. ∎

The log-concavity inequality.

Lemma 3.3. *Under the hypotheses of Theorem 3.1, for $i, j \ge 1$ and $\ell \ge 0$,*

$$f_i(\ell)\, f_j(\ell+1) \;\ge\; d_{\ell+1}\, f_{i+j}(\ell).$$

Proof. For a nonnegative finitely supported sequence u that is log-concave without internal zeros, $v_i = u_i + u_{i+1}$ has the same property, since $v_i^2 - v_{i-1}v_{i+1} = (u_i^2 - u_{i-1}u_{i+1}) + (u_{i+1}^2 - u_iu_{i+2}) + (u_iu_{i+1} - u_{i-1}u_{i+2})$ and the last term is nonnegative by log-concavity. The sequence $(0, \nu_1, \nu_2, \dots)$ has this property, and $f_i(\ell)$ is obtained from it by applying $u \mapsto v$ ℓ times. For such a sequence g, balanced products dominate: $g_ig_j \ge g_0g_{i+j}$ and $g_ig_{j+1} \ge g_1g_{i+j}$ for $i, j \ge 1$. Adding these with $g = f(\ell)$ and using $g_j + g_{j+1} = f_j(\ell+1)$ and $g_0 + g_1 = f_0(\ell+1) = d_{\ell+1}$ gives the claim. ∎

Lemma 3.4. *Under the hypotheses of Theorem 3.1, for all $j \ge 1$ and $\ell, a \ge 0$,*

$$
\begin{aligned}
A_0(\ell,a)B_{j-1}(\ell+1,a-1) + B_0(\ell,a)A_{j-1}(\ell+1,a-2) &\ge 4d_{\ell+1}A_j(\ell,a),\\
B_0(\ell,a)B_{j-1}(\ell+1,a-2) &\ge 4d_{\ell+1}B_j(\ell,a),\\
B_0(\ell,a)C_{j-1}(\ell+1,a-2) &\ge 4d_{\ell+1}C_j(\ell,a),
\end{aligned}
$$

with $W_{p,q}(\ell,a) = 0$ for $a < 0$.

Proof. Use $A_0(\ell,a) \ge af_1(\ell)$ and $B_0(\ell,a) \ge \binom{a}{2}f_2(\ell)$, expand the second factor, apply Lemma 3.3 to $f_1(\ell)f_n(\ell+1)$ and $f_2(\ell)f_n(\ell+1)$, and use $a\binom{a-1}{n} = (n+1)\binom{a}{n+1}$ and $\binom{a}{2}\binom{a-2}{n} = \binom{n+2}{2}\binom{a}{n+2}$. After division by $d_{\ell+1}$ the three left sides are bounded below by $\sum_m \gamma_m\binom{a}{m}f_m(\ell)$ with

$$\gamma_m = \binom{m+1}{2}\binom{m-j-2}{j-1}, \qquad \binom{m}{2}\binom{m-j-3}{j-1}, \qquad \binom{m}{2}\binom{m-j-2}{j},$$

while the right sides are $4\sum_m \gamma'_m \binom{a}{m} f_m(\ell)$ with $\gamma'_m = \binom{m-j-1}{j}$, $\binom{m-j-2}{j}$ and $\binom{m-j-1}{j+1}$. Where $\gamma'_m > 0$ the quotients are

$$\frac{jm(m+1)}{2(m-j-1)}, \qquad \frac{jm(m-1)}{2(m-j-2)}, \qquad \frac{(j+1)m(m-1)}{2(m-j-1)},$$

on $m \geq 2j+1$, $m \geq 2j+2$ and $m \geq 2j+2$ respectively. They increase with j for fixed m; at $j = 1$ their minima over integers are 5, 5 and 6. All quotients exceed 4, and $f_m(\ell) \geq 0$. ∎

Proof of Theorem 3.1. Let $\Omega = \{(\ell, a) : \ell, a \geq 0,\ \ell + a \leq N\}$, on which ω is supported. Read (3.1) as moves into the node $(\ell+1, a)$, each divided by $d_{\ell+1} > 0$: the straight move from (ℓ, a) with weight $a_{\ell+a} \geq 0$ on Ω, and for each j the three moves from $(\ell - j, a + 2j + 1)$, $(\ell - j, a + 2j + 2)$, $(\ell - j - 1, a + 2j + 2)$ with weights tA_j, B_j, C_j. Every move raises ℓ. Let $\mathcal{U}$ be the straight moves, $\mathcal{P}$ the moves with j even and $\mathcal{Q}$ the moves with j odd, all with nonnegative weights, so that $\omega = \beta + (\mathcal{U} + \mathcal{P} - \mathcal{Q})\omega$ with $\beta = \omega|_{\ell=0}$.

We claim $\mathcal{P}^2 \geq 4\mathcal{Q}$ entrywise. A move of $\mathcal{Q}$ with index j from $x = (\ell, a)$ is matched by the two-step paths in $\mathcal{P}^2$ that start with A_0 or B_0 from x (index 0, landing at $(\ell+1, a-1)$ or $(\ell+1, a-2)$, which lie in Ω) and continue with the move of the same type and index $j-1$ (even), as in Lemma 3.4: A_j by A_0 then B_{j-1} and B_0 then A_{j-1}; B_j by B_0 then B_{j-1}; C_j by B_0 then C_{j-1}. These paths end at the target of the matched move, carry the same power of t, and have the extra denominator $d_{\ell+1}$ of the intermediate node. Distinct moves of $\mathcal{Q}$ have distinct displacements, so their paths are distinct, and Lemma 3.4 gives the claim.

Now show by induction on the level that $\omega \geq 0$ and $\omega \geq \frac{1}{2}\mathcal{P}\omega$. Both hold at $\ell = 0$. If they hold below a node y at level $\ell + 1 \geq 1$, then, all weights being nonnegative, $(\mathcal{P}\omega)(y) \geq \frac{1}{2}(\mathcal{P}^2\omega)(y) \geq 2(\mathcal{Q}\omega)(y)$, and

$$\omega(y) = (\mathcal{U}\omega)(y) + (\mathcal{P}\omega)(y) - (\mathcal{Q}\omega)(y) \geq \tfrac{1}{2}(\mathcal{P}\omega)(y) \geq 0.$$

Hence all $\omega_{\ell,a} \geq 0$, so K_k has nonnegative coefficients in z, w, so $L^k_{mn} \geq 0$, $H^k_{mn} \geq 0$ and $g(m, n; k) \geq 0$. ∎

Remark 3.5. The argument is valid coefficientwise in t. With independent parameters in $yp_i = a_i p_{i+1} + u d_i p_i + v d_i p_{i-1}$, weighted homogeneity ($\deg y = \deg u = 1$, $\deg v = 2$) then shows that each linearization coefficient is a polynomial in (u, v) with nonnegative coefficients: for $D = m + n - k \geq 0$ it has the form $\sum_{0 \leq j \leq \lfloor D/2 \rfloor} h_j u^{D-2j} v^j$ with $h_j \geq 0$, and it vanishes for $D < 0$. Consequently every $g(m, n; k)$ of the array of Theorem 3.1, as a function of r, equals $\sum_j h_j r^j (1-r)^{D-2j}$; multiplying the j-th term by $(r + (1-r))^j = 1$ shows that its coefficients in the Bernstein basis of degree D are $\binom{D}{s}^{-1} \sum_{j \leq s \leq 2j} h_j \binom{j}{s-j} \geq 0$.

Corollary 3.6 *(Full positive Krawtchouk range).* *For every $N \geq 1$ and $0 < p_0 \leq 1/2$, Krawtchouk polynomials orthogonal for binomial weights $\binom{N}{j} p_0^{N-j} (1-p_0)^j$ have nonnegative linearization when normalized at $j = 0$.*

Proof. To fix the convention without relying on a named recurrence, put $r = p_0/(1-p_0) \in (0,1]$, and let $Y_1, \dots, Y_N$ be independent variables with values $1, -r$ of probabilities $p_0, 1-p_0$. Their mean is zero and $Y_i^2 = (1-r)Y_i + r$. The normalized elementary symmetric sums

$$K_i(j) = \binom{N}{i}^{-1} \sum_{|S|=i} \prod_{l \in S} Y_l$$

depend only on the number j of negative coordinates. They are orthogonal, because the expectation of two distinct squarefree monomials is zero: some independent mean-zero variable occurs once. Their weights in j are exactly the stated binomial weights. With $x = N^{-1} \sum_l Y_l = 1 - j/(N(1-p_0))$, multiplication by x gives

$$xK_i = \frac{N-i}{N} K_{i+1} + (1-r)\frac{i}{N} K_i + r\frac{i}{N} K_{i-1}.$$

Indeed, in multiplying the elementary sum by $\sum_l Y_l$, indices outside a subset give $(i+1)$ copies of each subset of size $i+1$; the linear part of Y_l^2 gives $(1-r)i$ copies of each subset of size i; and the constant part gives $r(N-i+1)$ copies of each subset of size $i-1$. Division by the binomial normalizers gives the displayed recurrence, including $i = N$. Its leading coefficient in x is positive, its degree is i, and its value at $j = 0$ is one. These properties identify the normalized orthogonal polynomials for the binomial measure. This is the array $K_{N,r}$; its profile $d_i = i/N$ has Newton coefficients $(1/N, 0, \dots, 0)$, so Theorem 3.1 applies. The range is sharp in this endpoint normalization: if $1/2 < p_0 < 1$, then $r = p_0/(1-p_0) > 1$ and $g(1,1;1) = b_1 = (1-r)/N < 0$. For $N = 1$, this coefficient follows from the terminal relation. ∎

Corollary 3.7 *(Geometric arrays). For every $N \geq 2$, $q > 1$, and $0 < r \leq 1$, the array $G_{N,q,r}$ of (2.4) has nonnegative linearization. In the normalization $R_0 = 1$, $R_1 = x$, it is outside the Krawtchouk recurrence class.*

Proof. Its increments are $d_i - d_{i-1} = q^{i-1}/[N]_q$, so its Newton coefficients $\nu_k = (q-1)^{k-1}/[N]_q$ are log-concave without internal zeros and Theorem 3.1 applies. For a Krawtchouk recurrence in this normalization the increments of $b_i + c_i$ are constant. Here their ratio is $q > 1$ and $N \geq 2$, so they are not constant. An affine change of the support transporting the same endpoint-normalized basis leaves the multiplication coefficients in the basis unchanged; it cannot remove this distinction. ∎

The arrays $G_{N,q,r}$ are the normalized recurrences of Koelink's construction [8, Theorems 3.1(ii) and 7.1] with $r = 1/p$; their polynomial interpretation is the dual q-Krawtchouk interpretation described after his Theorem 7.8. A proof of Corollary 3.7 by a finite algebra with positive multiplication, which needs only the geometric increments, is given in [13].

For some parameters both families come from distance-regular graphs. Dividing the intersection array of such a graph by its valency gives an array satisfying (2.1), and its linearization coefficients are positive multiples of the intersection numbers p_{mn}^k, hence nonnegative [3, 4]. For an integer $s \geq 2$, the Hamming graph $H(N,s)$ gives $K_{N,r}$ with $r = 1/(s-1)$; a dual polar graph of diameter N with classical parameters $(N, q, 0, q^e)$ gives $G_{N,q,r}$ with $r = q^{-e}$. Corollaries 3.6 and 3.7 cover the full parameter ranges.

4. Szwarc's quadratic system

Problem 2 of [20, §5] does not stop at the Krawtchouk recurrence. With his parameter written as p_0, Szwarc introduces the symmetric recurrence of length $2N+1$

$$yQ_{2n} = (N-n)Q_{2n+1} + nQ_{2n-1}, \qquad yQ_{2n-1} = (1-p_0)Q_{2n} + p_0 Q_{2n-2}, \tag{4.1}$$

records that $Q_{2n}(y)$ is the Krawtchouk polynomial $K_n(y^2)$ of his recurrence, states that the Q_n have nonnegative linearization for $p_0 \le 1/2$, and asks for a general criterion that would handle this case. His Theorem 3.3 for infinite symmetric systems does not apply. In the normalization (4.1) its monotonicity hypotheses hold, and its hypothesis $\alpha_n \le \gamma_n$ holds at odd indices exactly when $p_0 \le 1/2$; at an even index $2n$ it reads $n \le N-n$ and fails in the upper half. Some failure of this kind is forced in every finite system, whose terminal upward coefficient vanishes. We derive the positivity of (4.1) from Theorem 3.1 through a quadratic decomposition of symmetric finite systems.

Put $M = 2N$, $R_{2n}(t) = Q_{2n}(\sqrt{N}t)$ and $R_{2n-1}(t) = \sqrt{N}\,Q_{2n-1}(\sqrt{N}t)$. Substitution in (4.1) shows that the R_k satisfy (2.2), with N replaced by M, for the array

$$b_k = 0, \qquad (a_{2n}, c_{2n}) = \Big(\frac{N-n}{N}, \frac{n}{N}\Big), \qquad (a_{2n-1}, c_{2n-1}) = (1-p_0,\, p_0). \tag{4.2}$$

It satisfies (2.1) with M in place of N whenever $0 < p_0 < 1$. The two normalizations differ by positive factors and a positive dilation, so they have the same product coefficients up to positive factors.

Szwarc uses the quadratic transformation [20, §3.5] to pass from a symmetric system to its even part. In [17] a parity-split monotonicity criterion for symmetric infinite systems gives property (P) of the generalized Chebyshev polynomials and, through their even parts, Gasper's result for the Jacobi polynomials with $\alpha \ge \beta$ and $\alpha+\beta+1 \ge 0$; in [18] the same route gives property (P) of Askey–Wilson polynomials for a range of parameters. In both, positivity passes from a symmetric system to its even part. For finite systems the converse direction also works: positivity of a symmetric system follows from positivity of two nonsymmetric finite systems, both read off from its coefficients. Any criterion for nonsymmetric finite systems therefore becomes a criterion for symmetric ones. For a family of infinite systems a result of this kind is known: Kahler [7] solved Problem 1 of [20, §5] by showing that the generalized Chebyshev polynomials, the quadratic transforms of the Jacobi polynomials, have property (P) exactly for the parameters for which the Jacobi polynomials $R_n^{(\alpha,\beta)}$ have it. His result concerns that family; Theorem 4.1 below is a reduction for arbitrary finite symmetric systems.

Theorem 4.1 *(Quadratic decomposition).* *Let $N \ge 2$, and let a_k, c_k $(0 \le k \le 2N)$ with $b \equiv 0$ satisfy (2.1) with $2N$ in place of N. Put $a_k = c_k = 0$ for $k \notin [0, 2N]$, and define the even component*

$$\hat{a}_n = a_{2n}a_{2n+1}, \quad \hat{b}_n = c_{2n}a_{2n-1} + a_{2n}c_{2n+1}, \quad \hat{c}_n = c_{2n}c_{2n-1} \qquad (0 \le n \le N)$$

and the odd component

$$\check{a}_n = a_{2n+1}a_{2n+2}, \quad \check{b}_n = a_{2n+1}c_{2n+2} + c_{2n+1}a_{2n}, \quad \check{c}_n = c_{2n+1}c_{2n} \qquad (0 \le n < N).$$

After the affine normalization $(\hat a_n, \hat b_n, \hat c_n) \mapsto (\hat a_n, \hat b_n - \hat b_0, \hat c_n)/\hat a_0$, *and likewise for the odd component, these arrays satisfy* (2.1) *on the indices* $0, \dots, N$ *and* $0, \dots, N-1$. *If both have nonnegative linearization, so does the symmetric array. More precisely, with* $\hat g$ *and* $\check g$ *the product coefficients of the components and terms with indices out of range omitted,*

$$g(2a, 2b; 2c) \;=\; \hat g(a, b; c), \qquad g(2a, 2b+1; 2c+1) \;=\; a_{2a}\,\check g(a, b; c) + c_{2a}\,\check g(a-1, b; c),$$

the coefficients with $m+n+k$ *odd vanish, and* $g(2a+1, 2b+1; 2c) \int R_{2c}^2\, d\mu \;=\; g(2c, 2a+1; 2b+1) \int R_{2b+1}^2\, d\mu$. *Conversely, nonnegative linearization of the symmetric array implies that of its even component.*

Proof. Since $b \equiv 0$, the matrix J is conjugate to $-J$ by $\operatorname{diag}((-1)^i)$, which fixes e_0; hence the measure of Lemma 2.1 is symmetric and $R_k(-t) = (-1)^k R_k(t)$. By symmetry $R_{2n}(t) = P_n(t^2)$ and $R_{2n+1}(t) = tL_n(t^2)$ for polynomials P_n, L_n of degree n. Applying (2.2) twice gives, with $x = t^2$,

$$xP_n \;=\; \hat a_n P_{n+1} + \hat b_n P_n + \hat c_n P_{n-1}, \qquad xL_n \;=\; \check a_n L_{n+1} + \check b_n L_n + \check c_n L_{n-1}.$$

The row sums of the components equal $a_{2n} + c_{2n} = 1$ and $a_{2n+1} + c_{2n+1} = 1$; the coefficients $\hat a_n, \check a_n$ are positive below the last index and vanish there, and $\hat c_n, \check c_n > 0$ for $n \ge 1$. Hence the normalized arrays, which describe the same polynomials in the variable $(x - \hat b_0)/\hat a_0$ (respectively $(x - \check b_0)/\check a_0$), satisfy (2.1). The terminal relation $tR_{2N} = c_{2N} R_{2N-1}$ on the support Σ of the symmetric system, applied once and twice, gives the terminal relations of P on $\{t^2 : t \in \Sigma\}$ and of L on $\{t^2 : t \in \Sigma,\ t \ne 0\}$. These sets have $N+1$ and N points, because Σ is symmetric with $2N+1$ points and contains 0. By Lemma 2.1 they are the supports of the two components.

The identity $tR_{2n} = a_{2n} R_{2n+1} + c_{2n} R_{2n-1}$ gives $P_n = a_{2n} L_n + c_{2n} L_{n-1}$ on the support of L; for $n = N$ it is the terminal relation, with $a_{2N} = 0$. Hence, on $\Sigma \setminus \{0\}$,

$$R_{2a} R_{2b+1} \;=\; t\big(a_{2a} L_a L_b + c_{2a} L_{a-1} L_b\big) \;=\; \sum_c \big(a_{2a}\check g(a, b; c) + c_{2a}\check g(a-1, b; c)\big) R_{2c+1},$$

and both sides vanish at $t = 0$. Similarly $R_{2a} R_{2b} = (P_a P_b)(t^2) = \sum_c \hat g(a, b; c) R_{2c}$ on Σ. Uniqueness of expansions on Σ gives the displayed formulas. The vanishing for $m+n+k$ odd and the relation for two odd indices follow from the symmetry of the measure and from $g(m, n; k) \int R_k^2\, d\mu = \int R_m R_n R_k\, d\mu$. The converse holds because $\hat g(a, b; c) = g(2a, 2b; 2c)$. ∎

Combined with Theorem 3.1, the decomposition gives a parity-split criterion of the same shape as Szwarc's Theorem 3.3: separate conditions on the even-indexed and the odd-indexed coefficients.

Corollary 4.2 *(Parity-split criterion).* *Let* $N \ge 1$ *and let a symmetric array* ($b \equiv 0$) *on the indices* $0, \dots, 2N$ *satisfy* (2.1). *Suppose that*

1. *the odd downward coefficients are constant,* $c_{2n+1} = p_0$, *with* $0 < p_0 \le 1/2$;

2. *the even downward coefficients* $d_n = c_{2n}$ *have positive increments* $d_n - d_{n-1}$, $1 \le n \le N$, *forming a geometric progression with ratio* $q \ge 1$.

Then the array has nonnegative linearization. With $r = p_0/(1-p_0)$ *and* $N \ge 2$, *its normalized even and odd components are* (2.4) *with parameters* (N, q, r) *and* $(N-1, q, r/q)$, *respectively. For* $N = 1$ *the odd component consists only of* $L_0 = 1$ *and the array is* (4.2). *If* $1/2 < p_0 < 1$ *in (i) and (ii) holds, then* $g(2,2;2) < 0$.

Proof. For $N = 1$ the array is $a_0 = 1$, $c_1 = p_0$, $a_1 = 1-p_0$, $c_2 = 1$, with support $\{-1, 0, 1\}$ and $R_1 = t$, $R_2 = (t^2 - p_0)/(1-p_0)$. On the support $R_1^2 = (1-p_0)R_2 + p_0$, $R_1R_2 = R_1$ and $R_2^2 = \frac{p_0}{1-p_0} + \frac{1-2p_0}{1-p_0}R_2$, as evaluation at $t = 0, \pm 1$ shows; all coefficients are nonnegative for $p_0 \le 1/2$, and $g(2,2;2) < 0$ for $p_0 > 1/2$. Let $N \ge 2$. The even component has $\hat{a}_n = (1-d_n)(1-p_0)$, $\hat{b}_n = d_n(1-p_0) + (1-d_n)p_0$ and $\hat{c}_n = d_n p_0$, so $\hat{a}_0 = 1-p_0$, $\hat{b}_0 = p_0$, and its normalization is $(1-d_n, (1-r)d_n, rd_n)$. Its profile d_n has geometric increments, so its Newton coefficients are proportional to $(q-1)^{k-1}$ and log-concave without internal zeros, and $0 < r \le 1$. For the odd component, $\check{a}_n = (1-p_0)(1-d_{n+1})$, $\check{b}_n = (1-p_0)d_{n+1} + p_0(1-d_n)$ and $\check{c}_n = p_0 d_n$. Geometric increments give $d_{n+1} - d_1 = qd_n$, so the normalization is

$$\Big(\frac{1-d_{n+1}}{1-d_1},\ \frac{(q-r)d_n}{1-d_1},\ \frac{rd_n}{1-d_1}\Big).$$

Its profile $qd_n/(1-d_1)$ has geometric increments with ratio q, and its downward coefficients are $r/q \le 1$ times the profile. Theorem 3.1 applies to both components, and Theorem 4.1 concludes. Comparison with (2.4), using $d_n = [n]_q/[N]_q$ and $1 - d_1 = q[N-1]_q/[N]_q$, identifies the parameters. For $p_0 > 1/2$, $g(2,2;2) = \hat{g}(1,1;1)$ is the normalized diagonal coefficient $(1-r)d_1 < 0$ (for $N = 1$ see above). ∎

Theorem 4.3 *(Szwarc's quadratic system).* *Let* $N \ge 1$ *and* $0 < p_0 \le 1/2$. *The array* (4.2) *has a symmetric positive orthogonality measure on* $2N+1$ *points, and for all* $0 \le m, n \le 2N$

$$R_m R_n = \sum_{k=0}^{2N} g(m,n;k) R_k, \qquad g(m,n;k) \ge 0, \qquad \sum_k g(m,n;k) = 1$$

on that support. If $1/2 < p_0 < 1$, *then* $g(2,2;2) < 0$.

Proof. Lemma 2.1 gives a positive measure on $2N+1$ points, and it is symmetric because $b \equiv 0$ (proof of Theorem 4.1). The array (4.2) satisfies the hypotheses of Corollary 4.2 with $c_{2n+1} = p_0$ and $d_n = n/N$, that is, with $q = 1$, which gives nonnegative linearization for $p_0 \le 1/2$ and $g(2,2;2) < 0$ for $1/2 < p_0 < 1$. The coefficients sum to one because $R_k(1) = 1$ and 1 lies in the support, since $J\mathbf{1} = \mathbf{1}$. ∎

The even component of (4.2) is the Krawtchouk array $K_{N,r}$ with $r = p_0/(1-p_0)$, so that $Q_{2n}(y)$ is a Krawtchouk polynomial in y^2, as Szwarc notes, and its odd component is $K_{N-1,r}$. A second proof of Theorem 4.3, by a graded version of the algebra of [13], and explicit coefficient neighborhoods of (4.2) are given there.

5. How far can such criteria go?

A criterion valid for all N cannot leave a margin of fixed size around the Krawtchouk arrays. Let

$$\operatorname{dist}(\widetilde{a},\widetilde{b},\widetilde{c}) \;=\; \max_{0\le i\le N} \max\Bigl(\Bigl|\widetilde{a}_i - \frac{N-i}{N}\Bigr|, \Bigl|\widetilde{b}_i - \frac{(1-r)i}{N}\Bigr|, \Bigl|\widetilde{c}_i - \frac{ri}{N}\Bigr|\Bigr)$$

be the coefficient distance to $K_{N,r}$.

Lemma 5.1 *(Corner value).* *For $0 < r \le 1$, the array $K_{N,r}$ has $g(N,N;N) = (1-r)^N$.*

Proof. Use the model in the proof of Corollary 3.6, with $p_0 = r/(1+r)$. There $R_N = K_N = Y_1 \cdots Y_N$ and $K_k = \binom{N}{k}^{-1} \sum_{|S|=k} Y_S$, where $Y_S = \prod_{l\in S} Y_l$. Since $Y_l^2 = (1-r)Y_l + r$,

$$R_N^2 \;=\; \prod_{l=1}^{N} \bigl((1-r)Y_l + r\bigr) \;=\; \sum_{S} (1-r)^{|S|} r^{N-|S|} Y_S.$$

Both sides of $R_N^2 = \sum_k g(N,N;k) R_k$ depend only on the number j of negative coordinates, and the identity holds on the support, that is, for every j; so it holds as an identity of functions of $(Y_1,\dots,Y_N)$. The 2^N monomials Y_S are orthogonal for the product measure, with $\mathbb{E}[Y_S^2] = r^{|S|} > 0$, hence linearly independent. The monomial $Y_{\{1,\dots,N\}}$ occurs only in K_N, with coefficient one, which gives $g(N,N;N) = (1-r)^N$. ∎

Thus the corner coefficient vanishes at $r = 1$. The next proposition raises the top mass of the orthogonality measure.

Proposition 5.2 *(Raising the top mass).* *Let $N \ge 1$ and $0 < r < 1$. The measure of $K_{N,r}$ has nodes $x_n = 1 - (1+r)n/N$ and masses $w_n = \binom{N}{n} r^{N-n}/(1+r)^N$. For $\tau > 0$ let μ^τ be this measure with the mass at $x_0 = 1$ raised to $(1+\tau)w_0$. Then μ^τ has*

$$g(N,N;N) \;=\; \frac{(1+\tau)^2(I-w_0)+w_0}{(1+\tau)(h-w_0)+w_0}, \qquad I \;=\; \bigl(r(1-r)\bigr)^N, \quad h \;=\; r^N,$$

which is negative exactly when $\tau > \tau_N^* = \bigl(1-(1-r^2)^N\bigr)^{-1/2} - 1$. After the affine normalization $R_n(1) = 1$, $R_1 = x$, its recurrence coefficients differ from those of $K_{N,r}$ by at most $3\tau \max_n w_n$.

Proof. Write π_n and $S_n = \sum_{j<n} \pi_j$ for $K_{N,r}$, and put $\varepsilon = \tau w_0$. The polynomial $(1+\varepsilon S_n)R_n - \varepsilon \sum_{j<n} \pi_j R_j$ has degree n and equals 1 at x_0. For $\deg p < n$, its integral against p with respect to $\mu^\tau = \mu + \varepsilon\delta_{x_0}$ is $-\varepsilon p(x_0) + \varepsilon p(x_0) = 0$, because $\sum_{j<n} \pi_j R_j$ reproduces $p(x_0)$. Hence it is the polynomial R_n^τ of μ^τ.

On the support, dual orthogonality gives $\sum_{j\le N} \pi_j R_j(x) = \mathbf{1}_{x=x_0}/w_0$, and in particular $S_N + \pi_N = 1/w_0$. Therefore $R_N^\tau = (1+\tau)R_N - \tau \mathbf{1}_{x_0}$ on the support. By Lemma 5.1 and $\pi_N = r^{-N}$ (Lemma 2.1), $\int R_N^3\, d\mu = I$; also $\int R_N^2\, d\mu = h$. Hence

$$\int (R_N^\tau)^3 d\mu^\tau \;=\; (1+\tau)^3(I-w_0) + (1+\tau)w_0,$$

$$\int (R_N^\tau)^2 d\mu^\tau \;=\; (1+\tau)^2(h-w_0) + (1+\tau)w_0,$$

and their ratio is the stated formula. Since $r > r/(1+r) > r(1-r)$, we have $h > w_0 > I$. So the denominator is positive, and the numerator is negative exactly when $(1+\tau)^2 > w_0/(w_0 - I) = \big(1-(1-r^2)^N\big)^{-1}$.

For the coefficients, comparing leading coefficients and using detailed balance, $\pi_n a_n = \pi_{n+1} c_{n+1}$, for both measures gives

$$a_n^\tau = a_n \frac{1+\varepsilon S_n}{1+\varepsilon S_{n+1}}, \qquad c_n^\tau = c_n \frac{1+\varepsilon S_{n+1}}{1+\varepsilon S_n},$$

and the row sums stay 1. Since $\varepsilon\pi_n = \tau w_n$, each of a_n, b_n, c_n changes by at most τw_n. Put $A_n = 1+\varepsilon S_n$ and $t_n = \varepsilon\pi_n = \tau w_n$. Since $b_0^\tau = \varepsilon/(1+\varepsilon)$, the affine coordinate is $y = (1+\varepsilon)x - \varepsilon$, and the normalized coefficients are $\widetilde{a}_n = (1+\varepsilon)a_n^\tau$, $\widetilde{b}_n = (1+\varepsilon)b_n^\tau - \varepsilon$, $\widetilde{c}_n = (1+\varepsilon)c_n^\tau$. For $n \geq 1$ we have $A_n \geq 1+\varepsilon$, and

$$\begin{aligned}
\widetilde{a}_n - a_n &= \varepsilon a_n - (1+\varepsilon)a_n t_n/A_{n+1},\\
\widetilde{c}_n - c_n &= \varepsilon c_n + (1+\varepsilon)c_n t_n/A_n,\\
\widetilde{b}_n - b_n &= -\varepsilon(a_n+c_n) + (1+\varepsilon)t_n\big(a_n/A_{n+1} - c_n/A_n\big).
\end{aligned}$$

Because $a_n, c_n \geq 0$ and $a_n + c_n \leq 1$, each absolute change is at most $\varepsilon + t_n \leq 2\tau \max_j w_j$. Row zero is restored exactly. This proves, in particular, the stated bound $3\tau \max_j w_j$ for every $\tau > 0$. ∎

Corollary 5.3 *(Exponential upper bound for the stability radius).* *Let $0 < r < 1$, and let $\rho_N(r)$ be the supremum of the ρ such that every array satisfying (2.1), with $\widetilde{b}_i \geq 0$ for all i and with $\operatorname{dist}(\widetilde{a},\widetilde{b},\widetilde{c}) \leq \rho$, has property (P). If $(1-r^2)^N \leq \frac{1}{2}$ and $3(1-r^2)^N \leq (1-r)/N$, then*

$$\rho_N(r) \leq 3\tau_N^* \max_n w_n \leq 3(1-r^2)^N.$$

In particular $\rho_N(r) = O\big((1-r^2)^N\big)$ as $N \to \infty$, for every fixed $r \in (0,1)$.

Proof. Put $x = (1-r^2)^N$ and $W = \max_n w_n$. For $0 < x \leq \frac{1}{2}$,

$$(1+x)^2(1-x) - 1 = x(1-x-x^2) > 0,$$

so $\tau_N^* < x$. Choose $\tau \in (\tau_N^*, x)$. Proposition 5.2 gives a negative corner coefficient and full-array distance at most $3\tau W < 3x \leq (1-r)/N$. Thus $\widetilde{b}_n > 0$ for $n \geq 1$, while $\widetilde{b}_0 = 0$. The exact coefficient formulas and the positive affine scale give $\widetilde{a}_n > 0$ for $n < N$, $\widetilde{c}_n > 0$ for $n > 0$, and all the basic boundary and row-sum equalities. Letting $\tau \downarrow \tau_N^*$ proves $\rho_N(r) \leq 3\tau_N^* W \leq 3x$. For each fixed $r \in (0,1)$, both smallness conditions hold for all sufficiently large N, since $N(1-r^2)^N \to 0$. ∎

Thus, for every fixed $r \in (0,1)$, no set of arrays with property (P) contains coefficient balls of a positive radius independent of N around all $K_{N,r}$. Corollary 5.3 bounds $\rho_N(r)$ from above only; it does not determine its order. For each fixed N, $\rho_N(r) > 0$ for $0 < r < 1$: explicit perturbation inequalities give $\rho_N(r) \geq \frac{1}{2}(\min(r, 1-r)/(32N))^N$, a lower bound that decays faster than every exponential in N, and, for fixed $N \geq 2$, $\rho_N(r)$ is of exact

order $(1-r)^N$ as $r \to 1$; at the symmetric endpoint $r = 1$, criteria that admit polynomial deformation rules, or analytic rules of the kind specified there, must satisfy two first-order equalities. These results are proved in [13].

How sharp is log-concavity? The Krawtchouk arrays, with only $\nu_1 \neq 0$, and the arrays $G_{N,q,r}$, with equality in every log-concavity inequality, both lie on the boundary of the hypothesis of Theorem 3.1. Positivity of the Newton coefficients of the profile does not suffice. For $q > 1$ and $0 < \lambda < 1$ the combination $(1-\lambda)K_{N,r} + \lambda G_{N,q,r}$ is a stochastic array with constant ratio b_i/c_i and increments proportional to $1 + t\,q^{i-N}$, where $t = \lambda N(q-1)q^{N-1}/((1-\lambda)(q^N-1))$; its Newton coefficients are positive, and for $N \geq 3$ one has $\nu_2^2 < \nu_1\nu_3$ (Section 3). Exact computation, machine-verified in Lean, gives $g(N, N; N) < 0$ for this combination at $(r, N, q, t) = (4/5, 10, 10, 1)$, $(3/5, 20, 10, 2)$, $(1/2, 20, 10, 1)$, $(2/5, 60, 10, 2)$, $(1/3, 100, 10, 2)$, $(3/10, 200, 2, 5)$ and $(2/7, 200, 10, 5)$. In particular, at these parameters no sufficient criterion whose admitted set is convex, such as a system of linear inequalities in the coefficients, can admit both $K_{N,r}$ and $G_{N,q,r}$.

Log-concavity is not necessary either. For $0 < \mu < 1$ and $q > 1$ let $D(i) = (1-\mu)i + \mu[i]_q$, whose Newton coefficients are $\nu_1 = 1$ and $\nu_k = \mu(q-1)^{k-1}$ for $k \geq 2$; thus $\nu_1\nu_3/\nu_2^2 = 1/\mu$, and the other log-concavity inequalities hold with equality. For $\mu = \frac{1}{10}$ and $q = \frac{11}{10}$, the constant-ratio arrays with profile $d_i = D(i)/D(N)$ have property (P) for every $r \in (0, 1]$ and $N \in \{6, 8, 10, 12\}$ (machine-verified); in the exact computation, every coefficient $g(m, n; k)$ is a polynomial in r with nonnegative Bernstein coefficients. For $\mu = \frac{1}{100}$ and the same q, already $N = 6$ gives a negative coefficient $g(6, 6; 6)$ at $r = 1$. At $r = 1$ we searched coarse grids of μ and q, with float64 screening and high-precision confirmation for $N \leq 20$ and high-precision arithmetic for $20 \leq N \leq 80$. The smallest factor $1/\mu$ for which (P) failed was $10^{5/2}$ at $N = 4$, 10^2 for $6 \leq N \leq 30$, and $10^{3/2}$ at $N = 40, 60, 80$; scripts and outputs are in the companion code [14]. These are the smallest failures found, not proven thresholds: the observed failures lay in narrow windows of (μ, q) on the tested grids. Let

$$V_N = \inf\{\mu^{-1} : 0 < \mu < 1,\ q > 1, \text{ the array with profile } D/D(N) \text{ and } r = 1 \text{ fails (P)}\},$$

with $\inf \varnothing = \infty$.

6. Open problems

The exact boundary. Determine the boundary between log-concavity and positivity of the Newton coefficients of the profile: the set of profiles for which every constant-ratio array has property (P). In particular, decide whether the threshold V_N of Section 5 tends to 1 as $N \to \infty$. The examples of Section 5 suggest that when log-concavity fails, the failures of (P) are driven by an acceleration of the increments near the terminal index.

The stability radius. For fixed $0 < r < 1$, determine the order of $\rho_N(r)$ as $N \to \infty$; in particular, decide whether $\liminf_N \rho_N(r)^{1/N} > 0$. Corollary 5.3 gives $\limsup_N \rho_N(r)^{1/N} \leq 1 - r^2$.

Other ratios. Theorem 3.1 requires b_i/c_i to be constant. The companion paper [12] admits ratios c_i/d_i that increase with the index. On log-concave Newton profiles with b nondecreasing, it admits linear and convex quadratic ratios, ratios of any degree whose Newton coefficients are log-concave without internal zeros, and cubic ratios under a single inequality. Criteria for other nonconstant ratios are open.

Monotonicity-type criteria. Szwarc's Theorems 3.2 and 3.3 in [20] are inequality conditions of monotonicity type for infinite systems. By Theorem 4.1, any such criterion for nonsymmetric finite systems, valid including the terminal indices, gives one for symmetric finite systems. Inequality conditions of this type for all products except those whose pairwise index sums all exceed N are given in [13]; for the remaining, deep products, Theorem 3.1 and its extensions in [12] give shape conditions that control them.

AI-use disclosure

Claude Opus 5.5 (Anthropic), run in the Claude Code agent; GPT-6 Astra or GPT-6.1 Sol (OpenAI; which of the two was not recorded), run in the Codex agent; and GPT-6 Pro (OpenAI), used in ChatGPT, were used to develop the proofs, the exact checks, the Lean formalization and the text. The kernel recurrence and the operator inequality in the proof of Theorem 3.1 arose in consultations with GPT-6 Pro, which also reviewed the results. This AI review is not human peer review, and no independent human verification is claimed. The author set the research objective and is responsible for the mathematical claims. The full disclosure is in the supplement [16].

Code and formalization

All theorems, corollaries and propositions, and Lemmas 2.1 and 5.1, are formalized in Lean 4 with Mathlib in the companion code [14], whose axiom audit admits only `propext`, `Classical.choice` and `Quot.sound`; "machine-verified" refers only to these Lean theorems. The supplement [16] lists the Lean declaration for each result and remark, describes the Lean routes that differ from the written proofs, and lists what is not formalized: the comparisons with the literature and the numerical searches concerning V_N. The documentation [15] describes the exact-arithmetic diagnostics.

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